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Package with multiple compartments

Abstract

The package includes an inner package including a first panel defining a first compartment, the first panel including a first end that is at least partially heat sealed, the first end defining a first tear-point, a second panel defining a second compartment, a first interfacing edge of the first panel being connected to a second interfacing edge of the second panel, the second panel including a second end that is at least partially heat sealed, the second end defining a second tear-point, the first re-closeable seal and the second re-closeable seal being on a third end of the first compartment and a fourth end of the at least one second compartment, respectively.

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Background/Summary

BACKGROUND

Field

(1) Example embodiments generally relate to a multi-compartment package.

Description of Related Art

(2) Packages can be used to store, transport and sell consumer products. The consumer products may be more enjoyable to consume when fresh. The consumer products may need to be disposed of following consumption.

SUMMARY

(3) At least one example embodiment is directed toward a package.

(4) In at least one example embodiment, the package includes an inner package including a first panel defining a first compartment, the first panel including a first end that is at least partially heat sealed, the first end defining a first tear-point that is configured to allow a first portion of the first end to be removed to allow access to a first re-closeable seal, a second panel defining a second compartment, a first interfacing edge of the first panel being connected to a second interfacing edge of the second panel, the second panel including a second end that is at least partially heat sealed, the second end defining a second tear-point that is configured to allow a second portion of the second end to be removed to allow access to a second re-closeable seal, and the first re-closeable seal and the second re-closeable seal being on a third end of the first compartment and a fourth end of the at least one second compartment, respectively.

(5) In at least one example embodiment, the first panel and the second panel are configured to be folded onto each other.

(6) In at least one example embodiment, the first end defines a first slit and the second end defines a second slit, and the first slit and the second slit are configured to align once the first panel and the second panel are folded onto each other.

(7) In at least one example embodiment, the first compartment includes a first empty compartment that is empty, and the second compartment includes a first filled compartment containing a first set of pouches and a second filled compartment containing a second set of pouches.

(8) In at least one example embodiment, the first empty compartment is a different size than the first filled compartment and the second filled compartment.

(9) In at least one example embodiment, each of the first set of pouches and the second set of pouches includes a quantity of consumer product pouches that are to be consumed during a first expected duration of time based on an average consumption rate.

(10) In at least one example embodiment, the first expected duration of time is shorter than a second expected duration of time, the second expected duration of time being a time period upon which flavor scalping is expected to occur for the quantity of the consumer product pouches following an exposure to ambient air.

(11) In at least one example embodiment, the package further includes an outer package, wherein the inner package is inside the outer package.

(12) In at least one example embodiment, the package further includes an outer package, the inner

package being contained in the outer package, wherein the first end defines a first slit and the second end defines a second slit, the at least one first panel and the at least one second panel being folded onto each other such that the first slit is aligned with the second slit, and the outer package defines at least one third slit, the at least one third slit being aligned with the first slit and the second slit.

(13) In at least one example embodiment, at least one tear line is defined by the first interfacing edge and the second interfacing edge.

(14) In at least one example embodiment, the package further includes an adhesive on at least a portion of a major interfacing surface between each of the first panel and the second panel to hold the inner package in a folded configuration, the folded configuration including the first panel and the second panel being folded onto each other.

(15) In at least one example embodiment, the first panel includes a plurality of first panels that are subdivided by tear lines, the plurality of first panels defining a plurality of first compartments that are the same as the first compartment.

(16) In at least one example embodiment, the plurality of first panels and the second panel are folded onto each other in an accordion configuration, and the inner package is configured to allow at least one of a first end panel or a second end panel, of the plurality of first panels and the second panel, to be separated and extended from sides of the accordion configuration.

(17) In at least one example embodiment, at least one tear line is defined by the inner package, the at least one tear line being configured to allow at least one of the first end panel or the second end panel to be removed from a remainder of the inner package.

(18) In at least one example embodiment, the plurality of first panels and the second panel are made from a recyclable material.

(19) In at least one example embodiment, the plurality of first compartments include one or more empty compartments, and the second panel includes a plurality of second compartments that define a plurality of second compartments, each one of the plurality of second compartments being filled compartments containing sets of pouches of a consumer product.

(20) In at least one example embodiment, the first panel and the second panel are made from a recyclable material.

(21) In at least one example embodiment, the first panel and the second panel are made from a layer of material, the layer of material including at least one first polymer layer and a gas/moisture impermeable layer, the at least one first polymer layer forming an interior surface of the first compartment and the second compartment.

(22) In at least one example embodiment, the gas/moisture impermeable layer is a foil layer.

(23) In at least one example embodiment, the first panel and the second panel are made from a layer of material that includes at least one gas/moisture impermeable layer, and an oxygen scavenger is one of enclosed within the first compartment, enclosed within the second compartment, infused within the layer of material, or connected to or infused with an inner layer of the layer of material, the inner layer forming an interior surface of the first compartment and the second compartment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The various features and advantages of the non-limiting embodiments herein may become more apparent upon review of the detailed description in conjunction with the accompanying drawings. The accompanying drawings are merely provided for illustrative purposes and should not be interpreted to limit the scope of the claims. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. For purposes of clarity, various dimensions of the drawings

may have been exaggerated.

(2) FIG. 1 is an illustration of an inner package, in accordance with at least one example embodiment;

(3) FIG. 2 is an illustration of an inner package containing a consumer product, in accordance with at least one example embodiment;

(4) FIG. 3 is an illustration of an inner package with opened compartments, in accordance with at least one example embodiment;

(5) FIG. 4 is an illustration of an end of an opened compartment, in accordance with at least one example embodiment;

(6) FIG. 5 is an illustration of an inner package in a folded configuration, in accordance with at least one example embodiment;

(7) FIG. 6 is an illustration of an inner package being placed into an outer package, in accordance with at least one example embodiment;

(8) FIG. 7 is an illustration of an inner package being placed into another outer package, in accordance with at least one example embodiment;

(9) FIG. 8 is an illustration of a package in a sealed configuration, in accordance with at least one example embodiment;

(10) FIG. 9 is an illustration of the package of FIG. 8 in the sealed configuration, in accordance with at least one example embodiment;

(11) FIG. 10 is an illustration of another outer package in a sealed configuration, in accordance with at least one example embodiment;

(12) FIG. 11 is an illustration of another inner package, in accordance with at least one example embodiment;

(13) FIG. 12 is an illustration of another inner package, in accordance with at least one example embodiment;

(14) FIG. 13 is an illustration of another inner package, in accordance with at least one example embodiment;

(15) FIG. 14 is an illustration of another inner package, in accordance with at least one example embodiment;

(16) FIG. 15A is an illustration of a layer of material for forming an inner package, in accordance with at least one example embodiment;

(17) FIG. 15B is an illustration of a spool of a layer of material, in accordance with at least one example embodiment;

(18) FIG. 15C is an illustration of a re-closeable seal for an inner package, in accordance with at least one example embodiment;

(19) FIG. 15D is an illustration of a layer of material with portions of re-closeable seals connected to the layer of material, in accordance with at least one example embodiment;

(20) FIG. 15E is an illustration of a layer of material with portions of re-closeable seals connected to the layer of material, in accordance with at least one example embodiment;

(21) FIG. 15F is an illustration of a layer of material with a re-closeable seal connected to the layer of material, in accordance with at least one example embodiment.

(22) FIG. 15G is an illustration of an inner package made from the layer of material, in accordance with at least one example embodiment;

(23) FIG. 15H is an illustration of an inner package that has been filled with a consumer product, in accordance with at least one example embodiment;

(24) FIG. 15I is an illustration of an inner package in a folded configuration, in accordance with at least one example embodiment;

(25) FIG. 15J is an illustration of an inner package in use, where end panels have been separated and extended from a folded configuration, in accordance with at least one example embodiment;

(26) FIG. 15K is an illustration of end panels being removed from a folded configuration, where

additional panels have been separated from the folded configuration, in accordance with at least one example embodiment;

(27) FIG. 16 is an illustration of a cross-section of a layer of material for an inner package, in accordance with at least one example embodiment;

(28) FIG. 17 is an illustration of a cross-section of another layer of material for an inner package, in accordance with at least one example embodiment;

(29) FIG. 18 is an illustration of a cross-section of another layer of material for an inner package, in accordance with at least one example embodiment;

(30) FIG. 19 is an illustration of a cross-section of another layer of material for an inner package, in accordance with at least one example embodiment; and

(31) FIG. 20 is an illustration of a flowchart of a method of forming a package, in accordance with at least one example embodiment.

DETAILED DESCRIPTION

(32) Some detailed example embodiments are disclosed herein. However, specific structural and functional details disclosed herein are merely representative for purposes of describing example embodiments. Example embodiments may, however, be embodied in many alternate forms and should not be construed as limited to only the example embodiments set forth herein.

(33) Accordingly, while example embodiments are capable of various modifications and alternative forms, example embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit example embodiments to the particular forms disclosed, but to the contrary, example embodiments are to cover all modifications, equivalents, and alternatives thereof. Like numbers refer to like elements throughout the description of the figures.

(34) It should be understood that when an element or layer is referred to as being “on,” “connected to,” “coupled to,” or “covering” another element or layer, it may be directly on, connected to, coupled to, or covering the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout the specification. As used herein, the term “and/or” includes any and all combinations or sub-combinations of one or more of the associated listed items.

(35) It should be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, regions, layers and/or sections, these elements, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, region, layer, or section from another region, layer, or section. Thus, a first element, region, layer, or section discussed below could be termed a second element, region, layer, or section without departing from the teachings of example embodiments.

(36) Spatially relative terms (e.g., “beneath,” “below,” “lower,” “above,” “upper,” and the like) may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It should be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(37) The terminology used herein is for the purpose of describing various example embodiments only and is not intended to be limiting of example embodiments. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly

indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, and/or elements, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, and/or groups thereof.

(38) When the words “about” and “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value include a tolerance of $\pm 10\%$ around the stated numerical value, unless otherwise explicitly defined.

(39) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, including those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(40) Example embodiments are described herein with reference to cross-sectional illustrations that are schematic illustrations of idealized embodiments (and intermediate structures) of example embodiments. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, example embodiments should not be construed as limited to the shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

(41) FIG. 1 is an illustration of an inner package **100**, in accordance with an example embodiment.

(42) In at least one example embodiment, the inner package **100** includes panels **110**. In at least one example embodiment, the panels **110** are connected in a side-by-side configuration, as shown in FIG. 1. In at least one example embodiment, the inner package **100** includes a first panel **110a**, a second panel **110b** and a third panel **110c**. In at least one example embodiment, each of the panels **110** defines a compartment **115**. In another example embodiment, each panel **110** defines more than one compartment **115**. In at least one example embodiment, the first panel **110a** defines a first compartment **115a**, the second panel **110b** defines a second compartment **115b** and the third panel **110c** defines a third compartment **115c**.

(43) In at least one example embodiment, a re-closeable seal **120** is on an end **130** of each of the compartments **115**. In at least one example embodiment, the re-closeable seal **120** allows the end **130** of each of the compartments **115** to individually be opened and re-closed. In at least one example embodiment, the re-closeable seals **120** are collinearly aligned with each other across the panels **110** (as shown in FIG. 1).

(44) In at least one example embodiment, each of the panels **110** includes a heat sealed region **125** (explained in better detail in FIGS. 15A-15G). In at least one example embodiment, a border **135** exists between the heat sealed region **125** and each of the compartments **115**. In at least one example embodiment, the heat sealed region **125** surrounds (encompasses) each of the compartments **115**.

(45) In at least one example embodiment, a tear-point **105** is adjacent to the end **130** of each of the compartments **115**. In at least one example embodiment, the tear-point **105** is on one or both sides of each of the compartments **115**, near the re-closeable seal **120**. In at least one example embodiment, the tear-point **105** is aligned between the re-closeable seal **120** and the border **135** at the end **130** of the compartment **115** (see alignment line **140**).

(46) In at least one example embodiment, the tear-point **105** is a slit, a weakened or thinned area, a notch, combinations thereof, etc. In at least one example embodiment, the tear-point **105** is included on the inner package **100** to make the inner package **100** child-resistant. In at least one example embodiment, the tear-point **105** includes visual indicia indicating that the panel **110** of the inner package **100** is to be opened at this point. In at least one example embodiment, the visual indicia include a visible strip, a color, an arrow, text, a visible indentation, a change in texture or thickness, other visual indicators, or combinations thereof.

(47) In at least one example embodiment, the inner package **100** does not include a tear-point **105**. In at least one example embodiment, the inner package **100** is opened by being cut (for example, by using scissors), rather than by being torn open manually. In at least one example embodiment, the inner package **100** is made from a durable material that only allows for cutting, rather than tearing.

(48) In at least one example embodiment, an end **145** of one or more of the panels **110**, or all of the panels **110**, includes a slit **150**. In at least one example embodiment, the slit **150** is sized so that the inner package **100** is able to be hung on a hanger **800** (see FIG. **8**) of a consumer display rack. In at least one example embodiment, the slit **150** is adjacent to an end (upper end) **130** of the compartment **115**. In another example embodiment, the slit **150** is adjacent to a lower end **155** of the compartment **115**, or near a side edge **160** of the compartments **115**. In at least one example embodiment, the slit **150** traverses through the heat sealed region **125** of one or more of the panels **110**, or all of the panels **110** (as shown in FIG. **1**).

(49) In at least one example embodiment, a divider **165** exists along interfacing edges of each of the panels **110**. In at least one example embodiment, the interfacing edges are side edges of each of the panels **110**. In at least one example embodiment, the divider **165** is an embossed line or a fold line. In at least one example embodiment, the divider **165** is a perforated or a weakened line. In at least one example embodiment, the panels **110** are able to be separated along the divider **165** (see at least FIG. **13**). In at least one example embodiment, the divider **165** is bracketed by the heat sealed region **125**.

(50) FIG. **2** is an illustration of the inner package **100** containing a consumer product **200**, in accordance with an example embodiment.

(51) In at least one example embodiment, the consumer product **200** includes an oral product. In at least one example embodiment, the consumer product **200** includes one or more pouches **205**. In at least one example embodiment, the oral product is a tobacco product or a non-tobacco product. The oral product may include chewing tobacco, snus, moist snuff tobacco, dry snuff tobacco, or other smokeless tobacco and non-tobacco products for oral consumption. A smokeless tobacco product may include tobacco that is whole, shredded, cut, granulated, reconstituted, cured, aged, fermented, pasteurized, or otherwise processed. Tobacco may be present as portions of leaves, flowers, roots, stems, extracts, or any combination thereof. In at least one example embodiment, the oral product includes a tobacco extract, such as a tobacco-derived nicotine extract (e.g., white snus) alone or in combination with non-tobacco cellulosic materials (e.g., microcrystalline cellulose (MCC)). Where the oral product includes nicotine, with or without tobacco, the nicotine may be tobacco-derived nicotine or synthetic nicotine. The oral product may be provided loose, in a pouch, as a plug or twist, or in a desired shape. The oral product may be in the form of lozenges, chews, gums, pouches, sticks, tablets, pastilles, or any other suitable form.

(52) In at least one example embodiment, the oral product may have various ranges of moisture. In at least one example embodiment, the oral product is a dry oral product having a moisture content ranging from 5% by weight to 10% by weight. In at least one example embodiment, the oral product has a medium moisture content, such as a moisture content ranging from 20% by weight to 35% by weight. In at least one example embodiment, the oral product is a wet oral product having a moisture content ranging from 40% by weight to 55% by weight.

(53) In at least one example embodiment, the one or more pouches **205** include smokeless tobacco pouches, for example. The non-tobacco products may include herbal compositions, pharmaceutical medications, or other non-tobacco products. Herbs and other edible plants can be categorized generally as culinary herbs (e.g., thyme, lavender, rosemary, coriander, dill, mint, peppermint) and medicinal herbs (e.g., Dahlias, Cinchona, Foxglove, Meadowsweet, *Echinacea*, Elderberry, Willow bark). In some example embodiments, the non-tobacco products may include *cannabis* or *cannabis* extracts.

(54) In at least one example embodiment, one of the compartments **115** is left empty, and therefore does not include a consumer product. In at least one example embodiment, and as shown in FIG. **2**,

the third compartment **115c** is left empty. In at least one example embodiment, the third compartment **115c** can be used for spent (consumed) consumer products **15**. In at least one example embodiment, the third panel **110c** and the third compartment **115c** are a same size as the first panel **110a** and the first compartment **115a**, and a same size as the second panel **110b** and the second compartment **115b**. In another example embodiment, the third panel **110c** and the third compartment **115c** are a different size relative to the other panels **110** and compartments **115**. In at least one example embodiment, the third compartment **115c**, which is left empty, is larger than the other compartments **115**.

(55) In at least one example embodiment, the first compartment **115a** (first filled compartment) with the consumer product **200** can be opened, and the consumer product **200** can be consumed during a duration of time. Once the first compartment **115a** is opened, the consumer product **200** is exposed to ambient air, where the ambient air can contribute to an increased rate of “flavor scalping” (loss of flavor quality) of the consumer product **200**, due at least in part to oxygen degradation. In at least one example embodiment, the second compartment **115b** (second filled compartment) can remain sealed until the consumer product **200** in the first compartment **115a** is consumed, in order to maintain a freshness and reduce flavor scalping of the consumer product **200** in the second compartments **115b**.

(56) In at least one example embodiment, each of the compartments **115** includes a variety of different types and/or a differing amount of the consumer product **200**. In at least one example embodiment, the variety of different types includes products with differing flavors, strengths, forms (solid, semi-solid, gel, etc.), or combinations thereof. In at least one example embodiment, each one of the compartments **115** includes a same one type of the consumer product **200**, where each one of the compartments **115** contains a different type of the consumer product **200** and/or a different amount of the consumer product **200**.

(57) In at least one example embodiment, the inner package **100** includes one singular compartment containing the consumer product **200** (e.g., the first compartment **115a**), where the one singular compartment is connected to one additional compartment which is left empty (e.g., the third compartment **115c**). That is to say, in at least one example embodiment, the inner package **100** includes only two of the compartments **115**. In another example embodiment, the inner package **100** includes a singular compartment containing the consumer product **200** (e.g., the first compartment **115a**), where no other compartments are included in the inner package **100**.

(58) In at least one example embodiment, a quantity of the consumer product **200** within each of the compartments **115** is determined so as to mitigate or avoid flavor scalping. For instance, in at least one example embodiment, flavor scalping data in combination with an average consumption rate of the consumer product **200** can be used to determine a quantity of the consumer product **200** for each compartment **115**.

(59) In at least one example embodiment, a desiccant **230** can be included in the compartments **115** containing the consumer product **200** (e.g., compartments **115a/b**), where the desiccant **230** can also be included in any of the other inner packages described herein. In at least one example embodiment, the desiccant **230** is a hygroscopic substance capable of removing moisture within the compartments **115**. In at least one example embodiment, the desiccant **230** includes at least one of a silica gel, an activated charcoal, calcium sulfate, calcium chloride, a molecular sieve, or combinations thereof.

(60) In at least one example embodiment, each panel **110** includes a heat sealed end **210**, where a remainder **220a** of each panel **110** is not heat sealed. In at least one example embodiment, a heat sealed end **210c** (using the third panel **110c** as an example) includes the heat sealed region **125**, where a remainder **220c** of the third panel **110c** includes a non-heat sealed region **215**. In this example embodiment, a portion of the border **135** of the heat sealed region **115** is below the tear-point **105**.

(61) FIG. 3 is an illustration of the inner package **100** with some of the compartments **115** in an

opened configuration, in accordance with an example embodiment.

(62) In at least one example embodiment, and as shown in FIG. 3, an end **300** of the first panel **110a** is removed by tearing the end **300** from a remaining portion **305** of the first panel **110a**, so that the re-closeable seal **120** is accessible and able to be opened and closed to access or seal the first compartment **115a**.

(63) In at least one example embodiment, both the first compartment **115a** and the third compartment **115c** are torn open, allowing the consumer product **200** to be consumed from the first compartment **115a**. Following consumption, a spent (consumed) consumer product **310** can be discarded within the third compartment **115c** or another one of the compartments **115** that is empty (also see FIG. 15J).

(64) FIG. 4 is an illustration of the end **130** of the remainder **220** of the first panel **110a**, after the end **300** has been removed, in accordance with an example embodiment.

(65) In at least one example embodiment, the first panel **110a** (and similarly, all the panels **110**) are formed from a combination of a first sheet of material (first portion) **400** and a second sheet of material (second portion) **405**. In at least one example embodiment, and as described in relation to FIGS. 15A-15G, the first sheet of material **400** and the second sheet of material **405** can be joined at least in part by the heat sealed region **125**. In another example embodiment, the first sheet of material **400** and the second sheet of material **405** can be extruded or blown so that the sheets of material **400/405** are, in essence, one sheet of material with no seams. In at least one example embodiment, the sheets of material **400/405** are joined at edges by other means, besides heat sealing, or in addition to heat sealing. In at least one example embodiment, the other means of joining the sheets of material **400/405** include the use of an adhesive, stitching, folding, spot heat sealing (in combination or in lieu of heat sealed seams), application of a solvent causing the sheets of material **400/405** to partially dissolve and re-solidify to be joined at edges, etc. Other means of joining the sheets of material **400/405** are also contemplated.

(66) FIG. 5 is an illustration of the inner package **100** in a folded configuration **500**, in accordance with an example embodiment.

(67) In at least one example embodiment, the panels **110** are folded on top of each other, to lay flat or substantially flat on each other. In at least one example embodiment, the folded configuration **500** can be considered an “accordion-style configuration” or an “accordion configuration” (see the example of folding in FIG. 15H), as the panels **110** are folded onto each other.

(68) In at least one example embodiment, the folded configuration **500** has a footprint (as depicted in FIG. 5) that is substantially a size of one of the panels **110**.

(69) In at least one example embodiment, in the folded configuration **500**, the slit **150** of the panels **110** are able to align, allowing the inner package **100** to be hung.

(70) FIG. 6 is an illustration of the inner package **100** being placed into an outer package **600** to form a package **605**, in accordance with an example embodiment.

(71) In at least one example embodiment, the inner package **100** is folded into the folded configuration **500**, prior to the inner package **100** being slid into an open end of the outer package **600**.

(72) In at least one example embodiment, the inner package **100** is placed into the outer package **600**, where the slit **150** of each of the panels **110** aligns with a slit **610** in the outer package **600**.

(73) In this example embodiment, the outer package **600** and the inner package **100** are able to be hung on the hanger **800** (see FIG. 8).

(74) FIG. 7 is an illustration of the inner package **100** in the folded configuration **500** being placed into another outer package **700** to form a package **705**, in accordance with an example embodiment.

(75) In at least one example embodiment, the outer package **700** has a slit **710** that is able to be aligned with the slit **150** of the inner package **100**.

(76) FIG. 8 is an illustration of a front of the package **705** in a sealed (assembled) configuration, in

accordance with an example embodiment.

(77) In at least one example embodiment, and as shown in FIG. 8, the package **705** can be hung on the hanger **800**. In at least one example embodiment, the hanger **800** traverses through the slit **710** of the outer package **700** and the slit **150** of the inner package **100** to allow the package **705** to be hung. In another example embodiment, the hanger **800** traverses through the outer package **700** but does not traverse through the inner package **100**. In this example embodiment, the slit **150** in the inner package **100** (see FIG. 5) may not be included.

(78) FIG. 9 is an illustration of a back of the package **705** of FIG. 8 in the sealed (assembled) configuration, in accordance with an example embodiment.

(79) In at least one example embodiment, the slit **710** traverses through the front and back of the package **705**, as shown in FIGS. 8 and 9.

(80) FIG. 10 is an illustration of another package **1005**, in accordance with an example embodiment.

(81) In at least one example embodiment, the package **1005** includes an outer package **1000** with a tab **1015**. In at least one example embodiment, the tab **1015** includes a slit **1010**. In at least one example embodiment, the slit **1010** is used to hang the package **1005**.

(82) In at least one example embodiment, the outer packages (e.g., **600**, **700**, and **1000**) are made from cardboard, paperboard, paper, or combinations thereof. In at least one example embodiment, the outer packages are made from a recyclable material. In at least one example embodiment, the outer packages are made from a same material as the inner packages.

(83) FIG. 11 is an illustration of another inner package **1100**, in accordance with an example embodiment.

(84) In at least one example embodiment, the inner package **1100** includes panels **110** in a side-by-side configuration, where the panels **110** are also in an end-to-end configuration with a third panel **1105** connected at a bottom of the inner package **1100**. Specifically, in at least one example embodiment, the inner package **1100** includes a first panel **110a** with a first compartment **115a** and a second panel **110b** with a second compartment **115b**, where the third panel **1105** includes a third compartment **1110**. In at least one example embodiment, an interfacing end edge **1115** of the third panel **1105** is connected to interfacing end edges **1120** of the first panel **110a** and the second panel **110b**. In at least one example embodiment, the third panel **1105** and the third compartment **1110** are a different size relative to the panels **110** and compartments **115**. In at least one example embodiment, the third panel **1105** runs along a width of the panels **110**, where the third panel **1105** is connected to a lower interfacing edge of the panels **110**.

(85) In at least one example embodiment, at least one of the compartments **115/1110** of the inner package **1100** is left empty, and therefore does not include a consumer product **200**. In at least one example embodiment, the third compartment **1110** is left empty.

(86) In at least one example embodiment, the third compartment **1110** is larger (a greater volume), as compared to the other compartments **115**.

(87) In at least one example embodiment, each of the panels **110/1105** includes the tear-point **105**. In at least one example embodiment, and as shown in FIG. 11, some or all of the tear-points **105** may include a thinned area **1125** that assists in tearing open at least some of the panels **110/1105** to access the re-closeable seal **120** and a respective one of the compartments **115/1110**.

(88) In at least one example embodiment, the slit **150** is included near the upper end of the inner package **1100**. In at least one example embodiment, the inner package **1100** is able to be folded along fold line **1130**, so that the first panel **110a** and the second panel **110b** are folded on top of each other, and the third panel **1105** is folded along a mid-section of the third panel **1105**.

(89) FIG. 12 is an illustration of another inner package **1200**, in accordance with an example embodiment.

(90) In at least one example embodiment, the inner package **1200** includes panels **110** that number four or more. In at least one example embodiment, the inner package **1200** includes a fourth panel

110d with a fourth compartment **115d** that is left empty, such that the empty compartment **115d** does not include a consumer product **200**. In at least one example embodiment, the fourth compartment **115d** is a same size as the other compartments (**115a-c**). In at least one example embodiment, the fourth compartment **115d** is a different size relative to the other compartments (**115a-c**). In at least one example embodiment, the fourth compartment **115d** is a larger size (more volume), as compared to the other compartments (**115a-c**).

(91) In at least one example embodiment, some or all of the panels **110** include the heat sealed end **210**. In at least one example embodiment, a first portion **1205** of a border of the inner package **1200**, near the tear-points **105** on side edges of the inner package **1200**, can include the heat sealed regions **125**. In at least one example embodiment, the first portion **1205** is not heat sealed. In at least one example embodiment, a second portion **1210** of an outer border around outer edges of the compartments **115**, that is on a lower portion **1215** of the inner package **1200**, includes the heat sealed regions **125**. In at least one example embodiment, the second portion **1210** is not heat sealed. In at least one example embodiment, the divider **614** is a seam that joins the first sheet of material **400** and the second sheet of material **405** (FIGS. 4 and 15A) to form each of the panels **110**. In another example embodiment, the divider **165** is a seam that is formed during an extrusion or a blowing process. In at least one example embodiment, the divider **165** is bracketed by a heat sealed region. In at least one example embodiment, the divider **165** is a heat sealed region. It should be understood that, if the divider **165** is formed via heat sealing, any heat sealed portions **1220** of the divider **165** that traverse through the re-closeable seal **120** will cause the re-closeable seal **120** to become non-functional in the heat sealed portions **1220**. That is to say, the heat sealed portions **1220** will cause the re-closeable seal **120** to remain closed in these locations, such that the re-closeable seal **120** will be unable to be opened and re-closed in the heat sealed portions **1220**. (92) FIG. 13 is an illustration of another inner package **1300**, in accordance with an example embodiment.

(93) In at least one example embodiment, the inner package **1300** includes five or more panels **110**. In at least one example embodiment, at least two of the compartments **115** are left empty, and therefore do not contain a consumer product **200**. In at least one example embodiment, the inner package **1300** includes the fourth panel **110d** with the fourth compartment **115d** and a fifth panel **110e** with a fifth compartment **115e**, where these two compartments **115d/115e** are left empty.

(94) In at least one example embodiment, a number of the compartments that are left empty equal a number of the compartments that contain the consumer product **200**. In another example embodiment, a number of the compartments that are left empty are less than a number of the compartments that contain the consumer product **200**.

(95) In at least one example embodiment, some or all of the interfacing edges between the panels **110** include a divider **1305** that defines a tear-line. In at least one example embodiment, the divider **1305** is a perforated or weakened line that allows some or all of the panels **110** to be torn away from a remainder of the inner package **1300** (see FIG. 15K, as an example). In at least one example embodiment, only the dividers **1305** between the fourth panel **110d** and the fifth panel **110e** define a tear-line, thereby only allowing the fifth panel **110e** to be torn away from the inner package **1300**, whereas the remaining dividers **1305** are embossed lines or fold lines, that may or may not be bracketed or formed by heat sealing. In at least one example embodiment, each of the dividers **1305**, other than the divider **1305** between the third panel **110c** and the fourth panel **110d**, define a tear-line. In this example embodiment, the third panel **110c** and the fourth panel **110d** are configured to remain together, up to a point in time when the inner package **1300** is to be disposed of following use of the inner package **1300**.

(96) FIG. 14 is an illustration of another inner package **1400**, in accordance with an example embodiment.

(97) In at least one example embodiment, the inner package **1400** includes panels **110** in both a side-by-side and end-to-end configuration. In at least one example embodiment, the inner package

1400 includes the third panel **110c** in end-to-end configuration with only one other panel, such as the first panel **110a**.

(98) In at least one example embodiment, the third compartment **115c** is left empty, such that the third compartment **115c** does not contain the consumer product **200**. In at least one example embodiment, more than one of the compartment **115** is left empty.

(99) In at least one example embodiment, the inner package **1400** includes panels **110** that are only in the end-to-end configuration. For instance, in at least one example embodiment, the inner package **1400** includes the first panel **110a** and the third panel **110c**, and not the second panel **110b**.

(100) In at least one example embodiment, the slit **150** is included on a lower end of the third panel **110c**. In at least one example embodiment, the slits **150** each become aligned once the inner package **1400** is folded into a folded configuration. In at least one example embodiment, in order to fold the inner package **1400** into the folded configuration, the first panel **110a** and the second panel **110b** are folded over each other, and the third panel **110c** is folded up to also be folded on top of the first panel **110a** and the second panel **110b**.

(101) FIG. **15A** is an illustration of a layer of material **1500** for forming an inner package, in accordance with an example embodiment. FIG. **15B** is an illustration of a spool **1570** of the layer of material **1500**, in accordance with at least one example embodiment. FIG. **15C** is an illustration of the re-closeable seal **120** for an inner package, in accordance with at least one example embodiment.

(102) In at least one example embodiment, the layer of material **1500** is one single sheet that is folded down a fold line **1505**, so that the first sheet of material **400** and the second sheet of material **405** are one single sheet that is divided by the fold line **1505**. In at least one example embodiment, the fold line **1505** corresponds with a centerline **1595** that runs along a longitudinal length of the layer of material **1500**. In another example embodiment, the sheet of material **1500** is more than one sheet. In at least one example embodiment, the first sheet of material **400** is separate from the second sheet of material **405**, where the fold line **1505** is a seam that joins the first sheet of material **400** and the second sheet of material **405** to form the sheet of material **1500**. In at least one example embodiment, the fold line **1505** is a heat sealed line. In at least one example embodiment, the first sheet of material **400** and the second sheet of material **405** can be joined via an adhesive, stitching, folding, or by other means, in addition to or alternatively from heat sealing.

(103) In at least one example embodiment, the layer of material **1500** includes one or more pliable layers of material with an efficient water/vapor transmission rate that provides a barrier to protect against flavor scalping of the consumer product **200**, especially following an exposure of the consumer product to ambient air. In at least one example embodiment, a pliable nature of the layer of material **1500**, where the layer of material **1500** is capable of heat sealing and/or crimping, can ease manufacturing adjustments. For example, potential changes to a head-space (internal volume) within the compartments **115** (FIG. **15G**) can be adjusted by cutting and forming different sizes of the layer of material **1500**, where the use of heat sealing/crimping can further fine-tune (minimize) the head-space. In at least one example embodiment, the cutting and forming of the layer of material **1500**, and the heat-sealing and/or crimping of the layer of material **1500**, is accomplished to minimize the head-space in order to reduce oxygen and/or moisture exposure for the consumer product **200** in each of the compartments **115**. In at least one example embodiment, the layer of material **1500** can be used in bulk quantities and cut to needed sizes that may vary for an array of inner packages, where heating sealing and/or crimping can further adjust a size of the compartments **115** (FIG. **15G**). In at least one example embodiment, adjusting for a desired head-space to contain the consumer product **200** within the compartments **115** can, by itself, assist in maintaining freshness of the consumer product **200**. In at least one example embodiment, use of the layer of material **1500** can assist manufacturing efforts by being easy to work with in terms of being able to use vertical form, fill and seal (VFFS) machines that are able to quickly form and fill the inner packages made from the layer of material **1500**.

(104) As shown in FIG. 15B, and in at least one example embodiment, the layer of material **1500** can be wound around a mandrel **1575** to make the spool **1570**. In at least one example embodiment, an end **1545** of the layer of material **1500** can be pulled or unwound from the spool **1570**, prior to folding of the layer of material **1500** along the fold line **1505** (FIG. 15A). In at least one example embodiment, the re-closeable seal **120** can be formed from a first strip **1580** and a second strip **1585** of material, where the first strip **1580** and the second strip **1585** define interlocking grooves **1555** (FIG. 15C). In at least one example embodiment, the first strip **1580** and the second strip **1585** can be made from a same material as the layer of material **1500**.

(105) FIG. 15D is an illustration of the layer of material **1500** with portions of re-closeable seals **120** connected to the layer of material **1500**, in accordance with at least one example embodiment. FIG. 15E is an illustration of the layer of material **1500** with portions of re-closeable seals **120** connected to the layer of material **1500**, in accordance with at least one example embodiment. FIG. 15F is an illustration of the layer of material **1500** with the re-closeable seal **120** connected to the layer of material **1500**, in accordance with at least one example embodiment.

(106) In at least one example embodiment, the first strip **1580** and the second strip **1585** can be cut into sections and bonded onto the layer of material **1500** to form a pre-processed layer of material **1590** (FIG. 15D) that can later be formed into an inner package **1510** (FIG. 15G). In at least another example embodiment, the first strip **1580** and the second strip **1585** can remain uncut, or cut into lengthy sections (sections that can be formed into a complete inner package, or several inner packages), where the first strip **1580** and the second strip **1585** are bonded onto the layer of material **1500** to form a pre-processed layer of material **1597** (FIG. 15E) that can later be formed into the inner package **1200** (FIG. 12). In at least one example embodiment, the re-closeable seal **120** initially remains one singular element (e.g., the first strip **1580** and the second strip **1585** are not separated), and a lower side of the re-closeable seal **120** is first bonded to one side of the layer of material **1500** (as shown in FIG. 15F), and later during a further processing of the layer of material **1550** an upper side **120a** of the re-closeable seal **120** is then bonded to another side of the layer of material **1500** (e.g., the re-closeable seal **120** is bonded to the second sheet of material **405**). In at least one example embodiment, the re-closeable seal **120**, the first strip **1580** and/or the second strip **1585** are bonded onto the layer of material **1500** through heat sealing, the use of an adhesive, stitching, spot heat sealing (in combination or in lieu of heat sealing), an application of a solvent, etc. Other means of bonding are also contemplated.

(107) FIG. 15G is an illustration of the inner package **1510** that has been made from the pre-processed layer of material **1590** (using any of the pre-processing shown in FIGS. 15D-F), in accordance with an example embodiment.

(108) In at least one example embodiment, the first layer of material (or first portion) **400** and the second layer of material (or second portion) **405** are joined in order to form the inner package **1510**. In at least one example embodiment, the inner package **1510** includes a first (initial) heat sealed region **1515**, where the first sheet of material **400** and the second sheet of material **405** are joined, or partially joined, via the first heat sealed region **1515**, where this could for instance be a heat sealing (crimping) process. In at least one example embodiment, the first heat sealed region **1515** can be used, for example, to at least partially assist in defining the compartments **115**. In at least one example embodiment, the first heat sealed region **1515** can leave a top end **1520** of the inner package **1510** open and accessible for filling (see a border **1525** of the first heat sealed region **1515**, in FIG. 15G). In at least one example embodiment, an added heat sealing process at a second heat sealed region **1535** can be performed to close the top end **1520** of the inner package **1510**, once some or all of the compartments **115** are filled with the consumer product **200**.

(109) In at least one example embodiment, the inner package **1510** includes six or more panels **110**, where each panel **110** includes at least one compartment **115**. In at least one example embodiment, the inner package **1510** includes a fifth panel **110e** with a fifth compartment **115e** and a sixth panel **110f** with a sixth compartment **115f**. In at least one example embodiment, the fifth compartment

115e and the sixth compartment **115f** are left empty, while the other compartments **115** are filled with the consumer product **200**. In at least one example embodiment, none of the compartments **115** are left empty. In at least one example embodiment, half of the compartments **115** are left empty, or a different number of compartments **115** are left empty, while a remaining number of the compartments **115** are filled with the consumer product **200**.

(110) In at least one example embodiment, the re-closeable seals **120** and/or the tear-points **105** are stamped onto the first sheet of material **400** and/or the second sheet of material **405**, rather than pre-fabricated and then bonded onto the layer of material **1500** (as in FIG. **15D**). In at least one example embodiment, the stamping can occur prior to joining the first sheet of material **400** and the second sheet of material **405**, during joining, or after joining. In another example embodiment, the re-closeable seals **120** and/or tear-points **105** are pre-fabricated on the sheet of material **1500** by other means, prior to the sheet of material **1500** being processed or cut to form the inner package **1510**.

(111) In another example embodiment, in lieu of folding the layer of material **1500** (FIGS. **15A** and/or **15D-F**) to form the inner package **1510**, the inner package **1510** can be formed from a material that is fabricated through a combined process of extruding, molding and/or blowing to form the inner package **1510** and the compartments **115**. Other fabrication methods are contemplated.

(112) In at least one example embodiment, the divider **165** at an interface **1530** between a panel with a full compartment and a panel with a compartment left empty (e.g., the divider between the fourth panel **110d** and the fifth panel **110e**) is a non-perforated divider **165**, whereas the remaining dividers **1305** are perforated lines or weakened lines that act as tear-lines. In at least one example embodiment, all of the dividers define tear-lines, or some of the dividers define tear-lines.

(113) In at least one example embodiment, the dividers **1305** are tear-lines that allow some of all of the panels **110** to be removed from the inner package **1510** as the consumer product **200** within the inner package **1510** is consumed (as discussed in more detail in FIGS. **15J** and **15K**). In at least one example embodiment, the fourth panel **110d** and the fifth panel **110e** may not be intended or designed to be separated from each other, as the fourth panel **110d** and the fifth panel **110e** are intended to be discarded together (as discussed in more detail in FIGS. **15J** and **15K**).

(114) In at least one example embodiment, either the divider **1305** or the divider **165** is used at an interface between each one of the panels **110**. In at least one example embodiment, other combinations of the divider **1305** and the divider **165** are contemplated.

(115) In at least one example embodiment, the slit **150** is included on a side edge of some of all of the panels **110**, rather than a top end as shown in FIG. **15G**. In at least one example embodiment, the slit **150** is included on a lower end of some or all of the panels **110**. In at least one example embodiment, the slit **150** of some or all of the panels **110** will align, once the inner package **900** is in a folded configuration (see at least FIG. **15I**).

(116) In at least one example embodiment, each of the slits **150**, the notches **105** and/or perforations for the divider **1305** (to form the divider **1305** into a tear-line) can be formed following or during heat sealing, or formed at any point prior to heat sealing including during the pre-processing of the layer of material (FIGS. **15D-F**), before the pre-processing or immediately following the pre-processing.

(117) FIG. **15H** is an illustration of the inner package **1510** that has been at least partially filled with the consumer product **200**, in accordance with an example embodiment.

(118) In at least one example embodiment, a second (additional) heat sealed region **1535** is applied to the inner package **1510** to seal the inner package **1510**. In at least one example embodiment, the second heat sealed region **1535** can be applied to seal a top portion of some or all of the compartments **115**. In at least one example embodiment, the second heat sealed region **1535** can abut the first heat sealed region **1515**. In at least one example embodiment, the second heat sealed region **1535** is applied to the inner package **1510** once at least some of the compartments **115** are

filled with the consumer product **200**. In at least one example embodiment, the inner package **1510** includes only a single heat sealing process, where the heat sealing occurs in the first heat sealed region **1515** and the second heat sealed region **1535**. In at least one example embodiment, more than two heat sealing processes are performed on the inner package **1510**. In at least one example embodiment, the inner package **1510** is heat sealed as the inner package **1510** is at least partially filled with the consumer product **200**, or after the inner package **1510** is at least partially filled with the consumer product **200**.

(119) In at least one example embodiment, an adhesive **1540** is applied to at least a portion of a front and/or back surface of the panels **110**, at locations that will interface with other panels **110** when the inner package **1510** is folded. In at least one example embodiment, the adhesive **1540** is applied at spot locations. In at least one example embodiment, the adhesive **1540** is applied at ends of the panels **110**, or at various locations on a front and/or back of some or all of the panels **110**, as shown in FIG. **15G**. In at least one example embodiment, the adhesive **1540** is a hot melt glue, a gummy adhesive, a fugitive glue, a repositionable glue, a silicon adhesive, or another type of adhesive that is capable of holding the panels **110** while allowing the panels **110** to later be separated with relatively low effort. In at least one example embodiment, the adhesive **1540** is a low temperature adhesive that remains in a semi-solid form at room temperature, and remains in a semi-solid form at normal outdoor ambient temperatures that may include a relatively wide range of temperatures (from hot climates to cold climates).

(120) In at least one example embodiment, and as shown in FIG. **15H**, the inner package **1510** is folded into a folded configuration **1550** (see FIG. **15I**), or an “accordion-style” configuration. In at least one example embodiment, and as shown in FIG. **15I**, the inner package **1510** is folded so that each panel **110** is folded onto the adjacent panel **110**, to form the folded configuration **1550**.

(121) In at least one example embodiment, the adhesive **1540** maintains the inner package **1510** in the folded configuration **1550**, by holding the panels **110** of the inner package **1510** to each other. In at least one example embodiment, an adhesive is not applied to the inner package **1510** that is in the folded configuration **1550**. In at least one example embodiment, heat sealing is used in lieu of, or in combination with, the adhesive **1540**, in order to hold some or all of the panels **110** together in the folded configuration **1550**. In another example embodiment, the panels **110** of the inner package **1510** are held together by other means, such as an application of tape, glue that hardens (becomes solid after drying), or other structure suitable for holding the panels **110** of the inner package **1510** together. In another example embodiment, each panel **110** of the inner package **1510** lies on top of each other, following folding, without an application of an adhesive or other structure.

(122) FIG. **15I** is an illustration of the inner package **1510** in the folded configuration **1550**, in accordance with an example embodiment.

(123) In at least one example embodiment, each of the panels **110** is folded on top of each other, in this configuration. In at least one example embodiment, the slit **150** of each of the panels **110** aligns, once the inner package **1510** is in the folded configuration **901**.

(124) In at least one example embodiment, the aligned slits **150** of the inner package **1510** in the folded configuration **1550** can allow the inner package **1510** to be hung on personal items such as a key chain, backpack, fanny pack, bicycle, cellphone case, etc.

(125) FIG. **15J** is an illustration of the inner package **1510** in use, where end panels **1565** (e.g., the first panel **110a** and the sixth panel **110f**) have been separated and extended from the remaining panels **110** of the inner package **1510** that remain in the folded configuration **1550**, in accordance with an example embodiment.

(126) In at least one example embodiment, and as shown in FIG. **15J**, the first panel **110a** includes the consumer product **200** in the first compartment **115a**, where a top end of the first panel **110a** has been removed via the tear-point **105** to access the consumer product **200**. In at least one example embodiment, a top end of the sixth panel **110f** has also been removed to allow the

consumer product **200** to be stored in the sixth compartment **115f** after the consumer product **200** has been consumed. In at least one example embodiment, the re-closeable seals **120** can retain the contents of the sixth compartment **115f** while maintaining a freshness of the consumer product **200** in the first compartment **115a**.

(127) In at least one example embodiment, the end panels **1565** do not need to be separated and extended from the remainder of the inner package **1510** for the end panels **1565** to be used. That is to say, in at least one example embodiment, a top end of one or both of the end panels **1565** can be removed and the compartments **115** can be accessed and used, while all panels **110** of the inner package **1510** remain in the folded configuration **1550** (see FIG. **15I**). Following use of the end panels **1565** (e.g., when the consumer product **200** in the first compartment **115a** of the first panel **110a** is fully consumed and/or when the sixth compartment **115f** of the sixth panel **110f** is full), either or both of the end panels **1565** can then be separated and removed from a remainder of the inner package **1510**. In at least one example embodiment, once the end panels **1565** are removed from a remainder of the inner package **1510**, new end panels (e.g., the second panel **110b** and the fifth panel **110e**) can be opened at the top, and accessed and used, while remaining in the folded configuration **1550**.

(128) In at least one example embodiment, the aligned slits **150** of the inner package **1510** in the folded configuration **1550** can allow the inner package **1510** to be hung on personal items, while allowing one or both of the end panels **1565** to be separated and extended to facilitate a convenient use of the inner package **1510**.

(129) FIG. **15K** is an illustration of the end panels **1565** (e.g., the first panel **110a** and the sixth panel **110f**) being removed from the folded configuration **1550**, where additional panels (e.g., the second panel **110b** and the fifth panel **110e**) are being separated and extended from the remainder of the inner package **1510**, in accordance with an example embodiment.

(130) In at least one example embodiment, and as shown in FIG. **15K**, the first compartment **115a** is empty as the consumer product **200** has been depleted, and the spent (consumed) consumer product **310** is contained in the sixth compartment **115f**. In at least one example embodiment, the end panels **1565** are removed from a remainder of the inner package **1510** by using the dividers **1305** that define tear lines. In at least one example embodiment, once the second panel **110b** and the fifth panel **110e** have been separated from the folded configuration **1550**, a top end of the second panel **110b** and the fifth panel **110e** can be removed to access the second compartment **115b** and the fifth compartment **115e**.

(131) In at least one example embodiment, once the first panel **110a** and the sixth panel **110f** are removed from a remainder of the inner package **1510**, the second panel **110b** and the fifth panel **110e** become a new set of end panels. Once the consumer product **200** in the second compartment **115b** of the second panel **110b** is used, the second panel **110b** can be removed from a remainder of the inner package **1510**. In another example embodiment, after the consumer product **200** in the first compartment **115a** is consumed, the first panel **110a** can remain connected to the inner package **1510** so that the first compartment **115a** can be re-purposed and used to retain the spent consumer product **310**.

(132) In at least one example embodiment, because the second compartment **115b** remained sealed, prior to depletion of the first compartment **115a**, the consumer product **200** in the second compartment **115b** remains fresh, prior to consummation.

(133) In at least one example embodiment, the inner package **1510** can be used by continuing to separate and remove end panels, prior to an ultimate disposal of the inner package **1510**. In at least one example embodiment, a final empty compartment (e.g. the fifth compartment **115e**, as shown in FIG. **15H**) may be configured to remain connected to a final filled compartment (e.g., the fourth compartment **115d**, as shown in FIG. **15G**), so that the final two panels (e.g., the fourth panel **110d** and the fifth panel **110e**) can more easily be disposed of effectively, together.

(134) FIG. **16** is an illustration of a cross-section of a layer of material **1600** that can be used to

form any of an inner package, in accordance with an example embodiment.

(135) In at least one example embodiment, the layer of material **1600** is used to make any of the inner packages (e.g., the inner package **100**, **1100**, **1200**, **1300**, **1400** and **1510**) of the example embodiments. In at least one example embodiment, the layer of material **1600** can be used in a same manner as the layer of material **1500** described in relation to FIG. 15A.

(136) In the example embodiment, the layer of material **1600** is gas and moisture impermeable. In at least one example embodiment, the layer of material **1600** is a multi-layer structure that includes a gas/moisture impermeable layer **1615** bracketed by a first polymer layer **1610** and a second polymer layer **1620** that are made from low melting point polymers. In at least one example embodiment, the gas/moisture impermeable layer **1615** is a foil layer that includes a foil, a foil laminate, a composite material with at least one foil layer, an aluminum layer, an aluminum alloy layer, a pliable metalized film layer, or a combination of one or more of these. In an alternative embodiment, the gas/moisture impermeable layer **1615** is a polymer that includes an elastomer, plastic, vinyl, rubber, butyl rubber, polyethylene, ethylene vinyl alcohol (EVOH), or a composite of any of these materials, or a composite of pliable or semi-pliable polymer materials that are gas and moisture impermeable. In at least one example embodiment, the gas/moisture impermeable layer **1615** includes at least one cyclic olefin copolymer (COC). In at least one example embodiment, the layer of material **1600** only includes one of more layers of COC.

(137) A “gas impermeable” material is a material with an oxygen transmission rate (OTR) of less than 0.016 cm.sup.2/m.sup.2/day at test conditions of 73° F. with 0% relative humidity using American Society for Testing and Materials (ASTM) standard D-3985. A “moisture impermeable” material is a material with a moisture vapor transmission rate (MVTR) of less than 0.016 grams of water/m.sup.2/day at test conditions of 100° F. with 90% relative humidity using ASTM standard F-1249. A “gas/moisture impermeable layer” is a layer of material that has both the “gas impermeable” and “moisture impermeable” properties.

(138) In at least one example embodiment, the polymer layers **1610/1620** are made from a relatively low melting point polymer, relative to the gas/moisture impermeable layer **1615**, where the polymer layers **1610/1620** may be a same polymer material, or a different polymer material. In at least one example embodiment, the polymer layers **1610/1620** include biaxially-oriented polyethylene terephthalate (BOPET), polyethylene (PE), polypropylene, other similar low-melting point polymers or thermoplastics, or combinations thereof. The polymer layers **1610/1620** may, for instance, include a layer or layers of elastomers, plastics, or other pliable polymer layers that have a relatively low melting point, or combinations of these materials. In at least one example embodiment, the polymer layers **1610/1620** are made from a material that allows oxygen migration/diffusion through the polymer layers **1610/1620**, with an efficient water/vapor transmission rate, where the materials listed above offer such properties.

(139) In at least one example embodiment, the polymer layers **1610/1620** are made from a recyclable material. In at least one example embodiment, the polymer layers **1610/1620** are made from at least one of polyethylene terephthalate (PET), high-density polyethylene (HDPE), or polyvinyl chloride (PVC), each of which are recyclable. In at least one example embodiment, the inner package is made only from one or more layers of recyclable material.

(140) In at least one example embodiment, an oxygen scavenger **1630** is connected to or infused within at least one of the polymer layers **1610/1620**. In at least one example embodiment, the oxygen scavenger **1630** is connected to a first outer surface **1640** of the first polymer layer **1610**, infused within the first polymer layer **1610**, or both. In at least one example embodiment, the first polymer layer **1610** is used to make an interior (inner) surface of the compartments of the inner packages. In another example embodiment, the oxygen scavenger **1630** is connected to the first outer surface **1640** of the first polymer layer **1610** and/or connected to a second outer surface **1650** of the second polymer layer **1620**, or infused within either or both of the first polymer layer **1610** and the second polymer layer **1620**.

(141) In another embodiment, active elements of the oxygen scavenger **1630** are infused into at least one of the polymer layers **1610/1620**. In at least one example embodiment, the active elements of the oxygen scavenger **1630** include iron fragments. In at least one example embodiment, the iron fragments are able to bind with a sequester free-oxygen via an iron oxidation reaction. In at least one example embodiment, a light-sensitive activator is co-mingled or infused with the active elements of the oxygen scavenger **1630**. In at least one example embodiment, the active elements of the oxygen scavenger **1630** and the light-sensitive activator are homogenously mixed together within one or both of the polymer layers **1610/1620**.

(142) In at least one example embodiment, the oxygen scavenger **1630** may come in various forms, and may or may not be in direct contact with the consumer product **200** in the compartments of the inner packages. In at least one example embodiment, the oxygen scavenger **1630** is exposed to an interior of the compartments, once the inner packages are formed and sealed. In at least one example embodiment, the oxygen scavenger **1630** is able to form an oxygen concentration gradient with an interior of the compartments, once the inner packages are formed and sealed, even if the oxygen scavenger **1630** is not in direct contact with an interior of the compartments.

(143) In some examples, a quantity of the oxygen scavenger **1630** is connected to or infused within an interior (inner) layer of each of the compartments, such that the oxygen scavenger **1630** is able to remove an expected volumetric quantity of oxygen within the compartments, once the inner packages are formed. In at least one example embodiment, the oxygen scavenger **1630** is not directly contacting the first outer surface **1640**, but is instead infused within the first polymer layer **1610**, or is in another polymer layer between the first polymer layer **1610** and the gas/moisture impermeable layer **1615**, where the oxygen scavenger **1630** is able to create an oxygen gradient that draws free-oxygen across the first outer surface **1640** and into or through the first polymer layer **1610** to bind with and sequester the free-oxygen.

(144) In at least one example embodiment, the light-sensitive activator includes molecules that, when exposed to a light source, will create a free-radical that binds with free-oxygen in a reduction reaction, so that the free-oxygen will in turn bind with the elements of the oxygen scavenger **1630** to initiate (activate) a further reaction between the oxygen scavenger **1630** and additional free-oxygen.

(145) In some embodiments that involve the oxygen scavenger **1630** and the light-sensitive activator being connected to or infused within the layer of material **1600**, one or both surfaces of the layer of material **1600** can be shielded from a light source prior to a complete formation of the inner package. This may be accomplished by keeping the oxygen scavenger **1630** and/or the first polymer layer **1610** covered, stored in a roll, or stored in a light-depleted environment, in order to ensure that the light-sensitive activator is not exposed to a light source that would otherwise inadvertently initiate the reaction between the light-sensitive activator and free-oxygen prior to the formation of the inner package.

(146) In at least one example embodiment, the first outer surface **1640** of the layer of material **1600** forms an interior surface of the compartments of the inner package, where the oxygen scavenger **1630** is able to draw free-oxygen to or through the first outer surface **1640**, once the inner package is sealed and at least some of the compartments of the inner packages contain the consumer product **200**. In at least one example embodiment, the oxygen scavenger **1630** is, over time, able to reduce a concentration of free-oxygen (molecular oxygen) within a head-space of the compartments, thereby effectively creating or approximating a chemically-inert atmosphere within the compartments. In some example embodiments, the chemically-inert atmosphere is created even in the absence of any process that may remove air and/or free-oxygen from the compartments, or replace air and/or free-oxygen with an inert gas, prior to the inner package being sealed. In another embodiment, a process of removing air or free-oxygen from the compartments, or replacing air or free-oxygen from the compartments, may be used in lieu of using the oxygen scavenger **1630** in or on the layer of material **1600**, or used in conjunction with using the oxygen scavenger **1630** in or on the layer of

material **1600**.

(147) In at least one example embodiment, the light-sensitive activator initiates a reaction with free-oxygen by being exposed to a light source that is ultraviolet (UV) light. In another embodiment, the light-sensitive activator initiates a reaction with free-oxygen by being exposed to a light source that is visible light, or portions of the spectrum of visible light. In some embodiments, the light-sensitive activator is part of the oxygen scavenger **1630**, or co-mingled with the elements or materials of the oxygen scavenger **1630**.

(148) In at least one example embodiment, the layer of material **1600** includes the gas/moisture impermeable layer **1615**, where the polymer layers **1610/1620** have a relatively low melting point, in order to provide a pliable or semi-pliable material that is easy to work with, is impervious to oxygen/gas exchange, and/or can be conveniently joined and/or crimped (heat sealed) using relatively low temperature heating. In at least one example embodiment, heat sealing of the layer of material **1600** of the inner package is accomplished using an applied pressure of about 20-40 psi (138-276 kPa) for a duration of less than 1 minute, with an applied temperature of about 137-205° C., where the polymer layers **1610/1620** (described herein) are able to be heat sealed under these conditions. In another example embodiment, heat sealing of the inner package is accomplished using an applied pressure of about 30 psi (207 kPa) with an applied temperature of about 137-205° C., or about 170° C., where the polymer layers **1610/1620** (described herein) are able to be heat sealed under these conditions.

(149) In at least one example embodiment, a matrix **1635** is connected to at least a portion of the first outer surface **1640** of the first polymer layer **1610**. In at least one example embodiment, the matrix **1635** includes the oxygen scavenger **1630**, or the active elements of the oxygen scavenger. In at least one example embodiment, the matrix **1635** is made from a material that allows oxygen migration/diffusion, in order to allow the oxygen scavenger **1630** to draw free-oxygen into the matrix **1635** and bind with the free-oxygen. The materials for the matrix **1635**, described herein, allow for such migration/diffusion. In at least one example embodiment, the matrix **1635** is a polymer matrix (polymer material) or a thermoplastic. In another embodiment, the matrix **1635** is made from polyethylene. In at least one example embodiment, the polymer material of the matrix **1635** has a melting point temperature that is similar, or the same, as the melting point temperature of the first polymer layer **1610**. In at least one example embodiment, the difference in melting point temperature between the first polymer layer **1610** and the polymer material of the matrix **1635** is about 25° C. or less, or about 10° C. or less, or about 5° C. or less. In at least one example embodiment, the polymer material of the matrix **1635** is a same material as the material that forms the first polymer layer **1610**. In at least one example embodiment, a melting point temperature of the polymer material of the matrix **1635** is lower than a melting point temperature of the first polymer layer **1610**.

(150) In at least one example embodiment, the matrix **1635** and/or the first polymer layer **1610** includes the desiccant **230**, either in conjunction with the oxygen scavenger **1630** or in lieu of the oxygen scavenger **1630**.

(151) In another example embodiment, the layer of material **1600** is a single layer material, where the single layer is the gas/moisture impermeable layer **1615**. In another embodiment, the layer of material **1600** is a multi-layer material, where each of the layers is the gas/moisture impermeable layer **1615**. In at least one example embodiment, the layer of material **1600** is devoid of polymer layers, or devoid of low melting point outer layers. In at least one example embodiment, the layer of material **1600** that includes the gas/moisture impermeable layer **1615** works in conjunction with a reduced head-space of each of the compartments **115** (as discussed herein), in order to reduce oxygen and/or moisture exposure for the consumer product **200** in each of the compartments **115**. This in turn can reduce flavor scalping caused at least in part by oxygen and moisture degradation.

(152) FIG. **17** is an illustration of a cross-section of another layer of material **1700** for an inner package, in accordance with an example embodiment.

(153) In at least one example embodiment, the layer of material **1700** is used to make any of the inner packages (e.g., the inner package **100**, **1100**, **1200**, **1300**, **1400** and **1510**) of the example embodiments. In at least one example embodiment, the layer of material **1700** can be used in a same manner as the layer of material **1500** described in relation to FIG. 15A.

(154) In at least one example embodiment, the layer of material **1700** is a multi-layer material that includes the first polymer layer **1610** and the gas/moisture impermeable layer **1615**. In an embodiment, the layer of material **1700** can be joined via heat sealing, or other means, to form any of the inner packages, where the joining involves joining the first polymer layer **1610** together. For instance, the layer of material **1700** is assembled so that the first polymer layer **1610** forms an interior of the compartments of the inner packages, and the gas/moisture impermeable layer **1615** forms an exterior surface of the compartments of the inner packages. In at least one example embodiment, the oxygen scavenger **1630** is connected to the first outer surface **1640** of the first polymer layer **1610**, the oxygen scavenger **1630** is infused through the first polymer layer **1610**, or both. In at least one example embodiment, the light-sensitive activator is co-mingled or infused along with the active elements of the oxygen scavenger **1630** (as described in relation to the example embodiment of FIG. 16).

(155) In another example embodiment, the layer of material **1700** includes the first polymer layer **1610** that is devoid of other layers. In this example embodiment, the oxygen scavenger **1630** may be included on either or both surfaces of the first polymer layer **1610** and/or the oxygen scavenger **1630** may be infused within the first polymer layer **1610**.

(156) FIG. 18 is an illustration of a cross-section of another layer of material **1800** for an inner package, in accordance with an example embodiment.

(157) In at least one example embodiment, the layer of material **1800** is used to make any of the inner packages (e.g., the inner package **100**, **1100**, **1200**, **1300**, **1400** and **1510**) of the example embodiments. In at least one example embodiment, the layer of material **1800** can be used in a same manner as the layer of material **1500** described in relation to FIG. 15A.

(158) In at least one example embodiment, the layer of material **1800** includes several layers that form a gas/moisture impermeable layer **1820**, where the gas/moisture impermeable layer **1820** includes: a biaxially-oriented polyethylene terephthalate (BOPET) layer **1840**, a polyethylene (that may be a clear or translucent polyethylene, or PE) layer **1835** and a foil layer **1830**. In at least one example embodiment, the layer of material **1800** also includes an adhesive layer **1825** and a sealant layer **1810**. In at least one example embodiment, the sealant layer **1810** is a polyethylene-based sealant, a polyethylene terephthalate (PET)-based sealant, a resin-based sealant, or combinations of these. In at least one example embodiment, the resin-based sealant is an ionomer resin sealant, such as a Surlyn® sealant, where Surlyn® is a registered trademark of DowDuPont Inc. In at least one example embodiment, the sealant layer **1810** is a TOPAS® cyclic olefin copolymer (COC) resin sealant, such as a CXBR sealant, where TOPAS® and CXBR are registered trademarks of Bemis Company, Inc. In at least one example embodiment, the sealant layer **1810** is connected to the foil layer **1830** by the adhesive layer **1825**. In at least one example embodiment, the sealant layer **1810** has a lower melting point relative to the gas/moisture impermeable layer **1820**. In at least one example embodiment, the gas/moisture impermeable layer **1820** includes at least one COC layer. In at least one example embodiment, at least one COC layer is used in lieu of the BPOET layer **1840** and/or the PE layer **1835**.

(159) In another example embodiment, layer of material **1600** (FIG. 16) is the gas/moisture impermeable layer **1820**, where the second polymer layer **1620** is connected to the adhesive layer **1825**. In another example embodiment, the layer of material **1700** (FIG. 17) is the gas/moisture impermeable layer **1820**, where the gas/moisture impermeable layer **1615** is connected to the adhesive layer **1825**. In at least one example embodiment, the gas/moisture impermeable layer **1820** is only the foil layer **1830**.

(160) In at least one example embodiment, the sealant layer **1810** is made from a material that

allows oxygen migration/diffusion through the sealant layer **1810**, where the materials for the sealant layer **1810** listed above allow for such oxygen migration/diffusion.

(161) In at least one example embodiment, the adhesive layer **1825** is a silicone-based adhesive, a food-grade polymer, a food-grade epoxy, combinations of these materials, or other suitable adhesives used for the storage of food-grade and/or consumable items.

(162) In at least one example embodiment, the layer of materials **1800** includes the following layers listed with respective thicknesses: 12 microns of the BOPET layer **1840**, 13 microns of the PE layer **1835**, 8.9 microns of the foil layer **1830**, 2.0 microns of the adhesive layer **1825** and 50.8 microns of the sealant layer **1810**. In at least one example embodiment, the sealant layer **1810** is at least four times a thickness of each of the other layers of the layer of materials **1800** (e.g., the BOPET layer **1840**, the PE layer **1835**, the foil layer **1830**, and the adhesive layer **1825**). In at least one example embodiment, the sealant layer **1810** is thicker than each of the other layers of the layer of materials **1800** combined (e.g., thicker than the combination of the BOPET layer **1840**, the PE layer **1835**, the foil layer **1830**, and the adhesive layer **1825**).

(163) In at least one example embodiment, in order to join the layer of material **1800** to form the inner package of some of the example embodiments, or the compartments of the inner packages the sealant layer **1810** is joined, causing the sealant layer **1810** to form an interior surface of compartments.

(164) In at least one example embodiment, the oxygen scavenger **1630** is connected to an outer surface **1850** of the sealant layer **1810**, the oxygen scavenger is infused within the sealant layer **1810**, or both. In at least one example embodiment, the exterior surface of the inner package is the BOPET layer **1840**. In at least one example embodiment, the light-sensitive activator is co-mingled or infused along with the active elements of the oxygen scavenger **1630** (as described in relation to the example embodiment of FIG. 16).

(165) FIG. 19 is an illustration of a cross-section of another layer of material **1900** for an inner package, in accordance with at least one example embodiment.

(166) In at least one example embodiment, the layer of material **1900** includes a layer of cellulose **1910** between the polymer layers **1610/1620**. In at least one example embodiment, the layer of cellulose **1910** includes paper, cellulose fibers, wood pulp, grasses, other vegetable materials, or combinations thereof. In at least one example embodiment, the layer of material **1900** includes only the polymer layer **1610** and the layer of cellulose **1910** but not the polymer layer **1620**.

(167) FIG. 20 is an illustration of a flowchart of a method of forming the package **705**, in accordance with at least one example embodiment.

(168) In at least one example embodiment, step **S2000** of the method includes bonding the re-closeable seal **120**, or portions of the re-closeable seal **120** (the first strip **1580** and the second strip **1585** of the re-closeable seal **120**), onto the layer of material **1550**, as shown for instance in FIG. 15D, 15E or 15F.

(169) In at least one example embodiment, step **S2005** includes first heat sealing portions of the layer of material **1550** to each other to form the inner package **1510** with the compartments **115** (FIG. 15F). In at least one example embodiment, this step can include folding the layer of material **1500** along the centerline **1595** (see at least FIGS. 15D, 15E and 15F), and then at least partially heat sealing the first portion **400** of the layer of material **1500** to the second portion **405** of the layer of material **1500**, at least in the first heat sealed region **1515** (FIG. 15G) of the inner package **1510**.

(170) In at least one example embodiment, step **S2010** includes filling the compartments **115** with the consumer products **200** (FIG. 15G), while the compartments **155** remain at least partially opened. As explained herein, this step can fill some but not all of the compartments **115**.

(171) In at least one example embodiment, step **S2015** includes second heat sealing a top portion of the inner package **1510** to close the compartments **115** that contain the consumer products **200**. In at least one example embodiment, this step can include heat sealing in the second heat sealed region **1535** (FIG. 15G) to close the compartments **115**.

(172) In at least one example embodiment, step S2020 includes folding the panels **110** of the inner package **1510** onto each other (see the folded configuration **1550** of the inner package **1510** in FIG. **15I**), and placing the inner package **1510** into the outer package **700** to form the package **705** (FIG. **8**).

(173) Features of the example embodiments can be combined, and the features described in some of the example embodiments can be implemented in the other embodiments. In at least one example embodiment, the layer of materials described in FIGS. **16-18** can be used to make any of the inner packages (e.g., inner packages **100**, **1100**, **1200**, **1300**, **1400** and **1510**). In at least one example embodiment, features of the panels, compartments, re-closeable seals, use of heat sealing (crimping), materials of construction, consumer products, inner packages and outer packages can be implemented or combined within each of the example embodiments.

(174) Example embodiments have been disclosed herein, it should be understood that other variations may be possible. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

Claims

1. A package, comprising: an inner package including a first panel defining a first compartment, the first panel including a first end that is at least partially heat sealed, the first end defining a first tear-point that is configured to allow a first portion of the first end to be removed to allow access to a first re-closeable seal, a second panel defining at least one second compartment, a first interfacing edge of the first panel being connected to a second interfacing edge of the second panel, the second panel including a second end that is at least partially heat sealed, the second end defining a second tear-point that is configured to allow a second portion of the second end to be removed to allow access to a second re-closeable seal, the at least one second compartment including a first filled compartment containing a first set of pouches with a first quantity of consumer product pouches that are to be consumed during a first expected duration of time based on an average consumption rate, the first expected duration of time being shorter than a second expected duration of time, the second expected duration of time being a time period upon which flavor scalping is expected to occur for the first quantity of the consumer product pouches following an exposure to ambient air, and the first re-closeable seal and the second re-closeable seal being on a third end of the first compartment and a fourth end of the at least one second compartment, respectively.

2. The package of claim 1, wherein the first panel and the second panel are configured to be folded onto each other.

3. The package of claim 2, wherein the first end defines a first slit and the second end defines a second slit, and the first slit and the second slit are configured to align once the first panel and the second panel are folded onto each other.

4. The package of claim 1, wherein the first compartment includes a first empty compartment that is empty, and the at least one second compartment further includes a second filled compartment containing a second set of pouches.

5. The package of claim 4, wherein the first empty compartment is a different size than the first filled compartment and the second filled compartment.

6. The package of claim 4, wherein the second set of pouches also includes the first quantity of consumer product pouches.

7. The package of claim 1, further comprising: an outer package, wherein the inner package is inside the outer package.

8. The package of claim 1, wherein at least one tear line is defined by the first interfacing edge and the second interfacing edge.

9. The package of claim 1, further comprising: an adhesive on at least a portion of a major

interfacing surface between each of the first panel and the second panel to hold the inner package in a folded configuration, the folded configuration including the first panel and the second panel being folded onto each other.

10. The package of claim 1, wherein the first panel includes a plurality of first panels that are subdivided by tear lines, the plurality of first panels defining a plurality of first compartments that are the same as the first compartment.

11. The package of claim 10, wherein the plurality of first panels and the second panel are folded onto each other in an accordion configuration, and the inner package is configured to allow at least one of a first end panel or a second end panel, of the plurality of first panels and the second panel, to be separated and extended from sides of the accordion configuration.

12. The package of claim 11, wherein at least one tear line is defined by the inner package, the at least one tear line being configured to allow at least one of the first end panel or the second end panel to be removed from a remainder of the inner package.

13. The package of claim 12, wherein the plurality of first panels and the second panel are made from a recyclable material.

14. The package of claim 11, wherein the plurality of first compartments include one or more empty compartments, and the second panel includes a plurality of second panels that define a plurality of second compartments, each one of the plurality of second compartments being filled compartments containing sets of pouches of a consumer product.

15. The package of claim 1, wherein the first panel and the second panel are made from a recyclable material.

16. The package of claim 1, wherein the first panel and the second panel are made from a layer of material, the layer of material including at least one first polymer layer and a gas/moisture impermeable layer, the at least one first polymer layer forming an interior surface of the first compartment and the at least one second compartment.

17. The package of claim 16, wherein the gas/moisture impermeable layer is a foil layer.

18. The package of claim 1, wherein the first panel and the second panel are made from a layer of material that includes at least one gas/moisture impermeable layer, and an oxygen scavenger is one of enclosed within the first compartment, enclosed within the at least one second compartment, infused within the layer of material, or connected to or infused with an inner layer of the layer of material, the inner layer forming an interior surface of the first compartment and the at least one second compartment.

19. A package, comprising: an inner package including a first panel defining a first compartment, the first panel including a first end that is at least partially heat sealed, the first end defining a first tear-point that is configured to allow a first portion of the first end to be removed to allow access to a first re-closeable seal, a second panel defining at least one second compartment, a first interfacing edge of the first panel being connected to a second interfacing edge of the second panel, the second panel including a second end that is at least partially heat sealed, the second end defining a second tear-point that is configured to allow a second portion of the second end to be removed to allow access to a second re-closeable seal; and an outer package, the inner package being contained in the outer package, wherein the first end defines a first slit and the second end defines a second slit, the first panel and the second panel being folded onto each other such that the first slit is aligned with the second slit, and the outer package defines at least one third slit, the at least one third slit being aligned with the first slit and the second slit, the first re-closeable seal and the second re-closeable seal being on a third end of the first compartment and a fourth end of the at least one second compartment, respectively.
